MediChain

Patient records. Coordinated care. Visible access history.

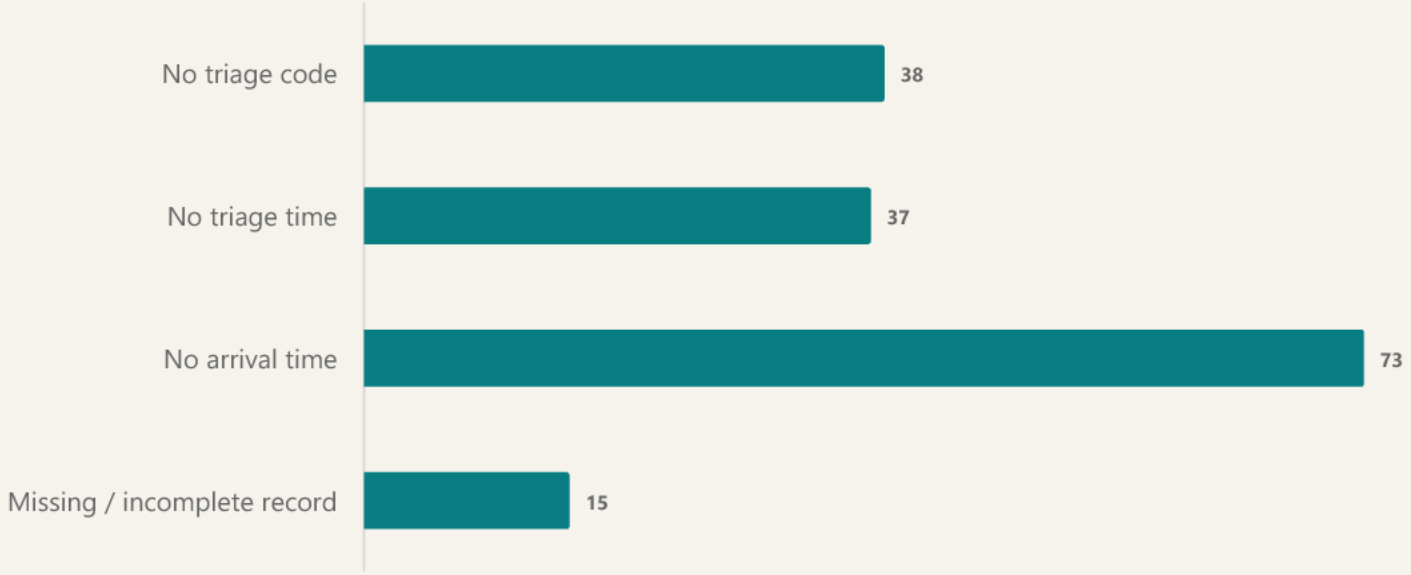
Patients need their care team to find the right information, while knowing who accessed their medical records and why.

A patient portal and role-specific clinical workspaces for routine care and controlled emergency access.

SA policy context: integrated records and a common patient identifier
National Department of Health, Digital Health Strategy 2019–2024

Problem evidence from South Africa

When records are missing or incomplete, the next clinician lacks reliable context and the patient cannot reconstruct their care.

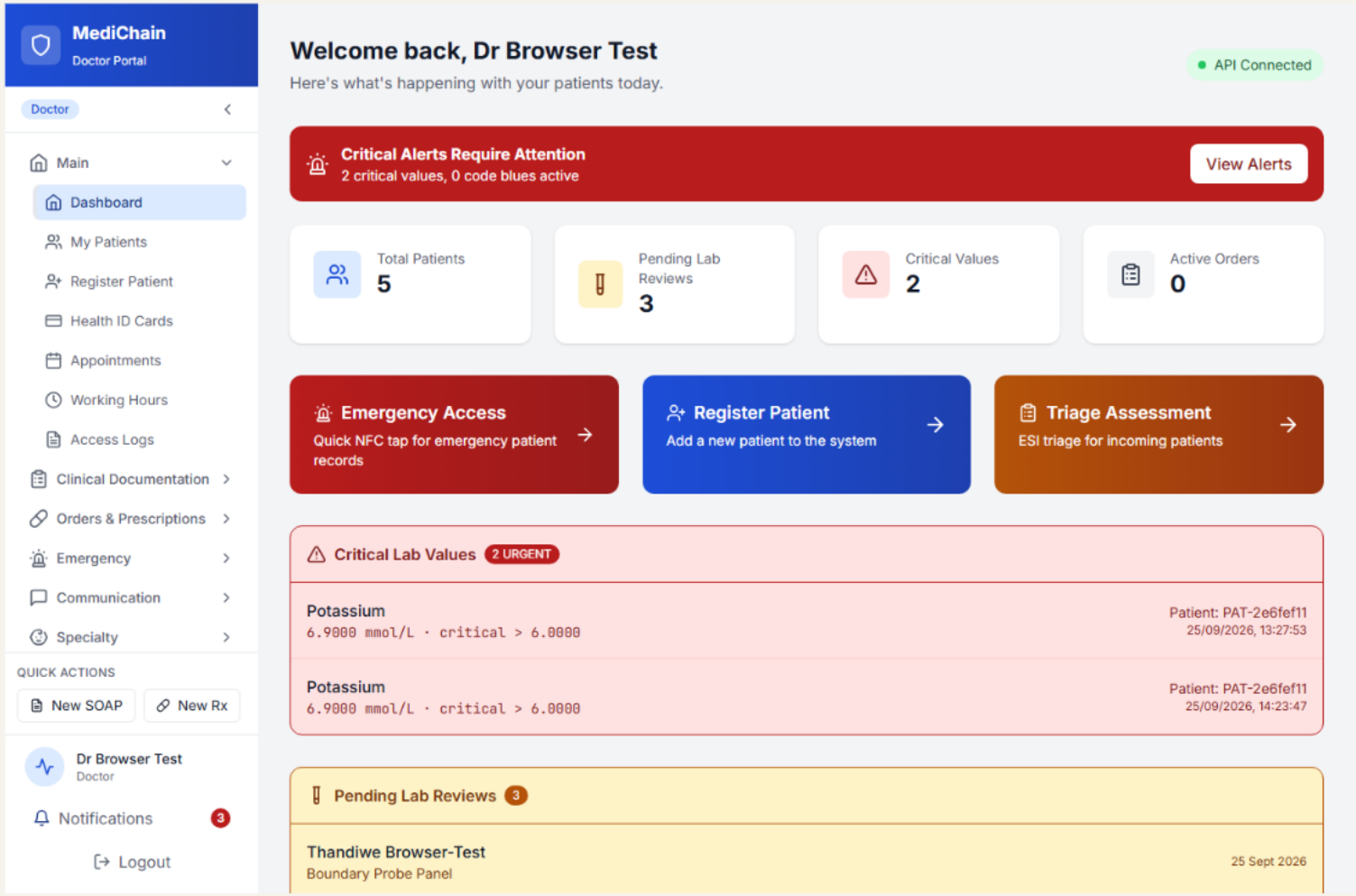


15%

Missing or incomplete records in the study's reviewed trauma cases.

Historical, single-hospital evidence: Khayelitsha, July 2012 records; study published 2016. Percentages overlap. This is not a current national rate or a MediChain outcome.

The solution in the application



Care team workspace

Patient chart and clinical notes
Orders and lab review
Prescriptions and dispensing
Controlled emergency access

Patient portal

Own records, medicines, consent and access history.

Seven roles, one patient care journey

Patient	View records and results, manage consent, review access history.
Doctor	Document consultations, order tests, prescribe and review results.
Nurse	Record vitals, triage, medication administration and shift handover.
Lab technician	Track specimens, enter results and escalate critical values.
Pharmacist	Review prescriptions, check drug safety and record dispensing.
Paramedic	Use controlled emergency access and record ambulance handover.
Administrator	Manage staff, approved devices, identity reviews and audit oversight.

A patient’s first visit and follow-up

01 Register and activate

Clinic registers the patient. Planned first-phone activation checks an ID, liveness and a matching face.

02 Receive care

Nurse records observations. Doctor reviews the chart, documents care and places orders.

03 Complete the orders

Lab technician returns results. Pharmacist reviews and dispenses the prescription.

04 Review and stay informed

Patient views records and medicines, manages consent and checks who accessed their data.

Emergency path: authorised responder + approved device + reason + time-limited access

First-phone identity verification

Implementation in progress

1. Capture evidence	Capture the ID and a live face in a short-lived session.
2. Check locally	OCR and document rules, liveness, then 1:1 face comparison.
3. Bind the account	Link the approved account to the correct patient medical ID.
4. Handle uncertainty	Route inconclusive checks to authorised manual review.

No external identity AI. No government-register claim. A medical ID does not grant access.

Architecture and patient-data protection

Application foundation

React patient and staff portals
Rust / Actix API
PostgreSQL clinical records
Encrypted document storage

Server checks identity, role, patient relationship and access purpose.
Required audit failure blocks disclosure.

Security work in progress

Private identity worker: local OCR, liveness and 1:1 face matching.
Inconclusive checks go to review.

Hash-chained audit + Merkle proofs
Encrypted S3 Object Lock backups
Optional RFC 3161 timestamps

Minimal evidence retention. No external identity AI. Medical ID is not a password.

Data privacy policies

Collection and purpose	Collect only information needed for care and identity checks.
Access and transparency	Check identity, role, patient relationship and access purpose.
Biometric evidence	Use restricted, encrypted temporary storage and defined deletion.
Patient safeguards	Show access history and provide a route to correct disputed details.
Retention and incidents	Confirm retention rules, processor agreements and incident duties.

Policy implementation and verification continue. Prototype uses synthetic data.

Competitive analysis

OpenMRS

Patient registration, appointments, queues, wards and clinical records.
Established EMR platform.

Bahmni

Integrated hospital workflows with EMR and laboratory systems.
Can operate at a hospital site.

MediChain

Patient-visible access history, role-specific care workflows and controlled emergency access.
Identity proofing in progress.

Our pilot must test whether this combination is easier to use and adopt.
No claim that existing platforms lack audit, privacy or patient-facing features.

Pilot adoption and a sustainable service

Initial customer

A clinic partner with a named clinical lead.
Start with one care workflow and synthetic training before an approved patient pilot.

Position among alternatives

Paper files: familiar local workflow.
Existing record systems: clinical operations.
MediChain focus: patient visibility linked to role-specific care and emergency access.

Proposed business model

Facility subscription plus onboarding and support. Validate pricing with clinics.
Ask: a supervised pilot partnership.

Readiness and pilot measures

TRL 3 framing: proof of concept.
Clinical validation remains outstanding.
Measure retrieval time, workflow completion and patient understanding of access history.

Readiness, demonstration and pilot ask

Proof of concept

Local sign-in and session restoration verified.
Clinical workflows need pilot validation.
Identity and audit upgrades remain in progress.

The demonstration

One synthetic patient: clinician record, patient view, access history.
Separate submission video must be under 90 seconds.

Ask: a clinic partner for supervised workflow evaluation.